PROGRAM – 3

Aim: Write a program implementing basic arithmetic operations of a simple calculator.

Program

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    android:padding="20dp"
    tools:context=".MainActivity">
    <EditText
        android:id="@+id/number1"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Number 01"
        android:inputType="numberDecimal"/>
    <EditText
        android:id="@+id/number2"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="10dp"
        android:hint="Enter Number 02"
        android:inputType="numberDecimal"/>
    <TextView
        android:id="@+id/result_text"
```

```
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="20dp"
    android:textColor="@color/black"
    android:textSize="175sp"
    android:textStyle="bold"/>
<Button
    android:id="@+id/add_btn"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="10dp"
    android:text="+"/>
<Button
    android:id="@+id/sub_btn"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="10dp"
    android:text="-"/>
<Button
    android:id="@+id/mul_btn"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="10dp"
    android:text="x"/>
<Button
    android:id="@+id/div_btn"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="10dp"
```

```
        android:text="/"/>
    <Button
        android:id="@+id/clear_btn"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="10dp"
        android:text="clear"/>
</LinearLayout>
```

MainActivity.java

```
package com.example.calculator;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {
EditText number1,number2;
TextView result_text;
Button add_btn,sub_btn,mul_btn,div_btn,clear_btn;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        number1=findViewById(R.id.number1);
```

```
number2=findViewById(R.id.number2);
result_text=findViewById(R.id.result_text);
add_btn=findViewById(R.id.add_btn);
sub_btn=findViewById(R.id.sub_btn);
mul_btn=findViewById(R.id.mul_btn);
div_btn=findViewById(R.id.div_btn);
clear_btn=findViewById(R.id.clear_btn);

 add_btn.setOnClickListener(new View.OnClickListener() {
    @Override
     public void onClick(View view) {
       String number1_text=number1.getText().toString();
       String number2_text=number2.getText().toString();
       int num1=Integer.parseInt(number1_text);
       int num2=Integer.parseInt(number2_text);
       float sum=num1+num2;
       result_text.setText("The addition of the two number is:"+sum);

       Toast.makeText(getApplicationContext(),"The addition of the two numbers
              is:"+sum, Toast.LENGTH_SHORT).show();
   }
});

sub_btn.setOnClickListener(new View.OnClickListener(){
  @Override
   public void onClick(View view){
       String number1_text=number1.getText().toString();
       String number2_text=number2.getText().toString();
       int num1=Integer.parseInt(number1_text);
```

```
        int num2=Integer.parseInt(number2_text);
        float sub=num1-num2;
        result_text.setText("The subtraction of the two number is:"+sub);

        Toast.makeText(getApplicationContext(),"The subtraction of the two numbers
              is:"+sub,Toast.LENGTH_SHORT).show();
    }
});

mul_btn.setOnClickListener(new View.OnClickListener(){
    @Override
    public void onClick(View view){
        String number1_text=number1.getText().toString();
        String number2_text=number2.getText().toString();
        int num1=Integer.parseInt(number1_text);
        int num2=Integer.parseInt(number2_text);
        float mul=num1*num2;
        result_text.setText("The multiplication of the two number is:"+mul);

        Toast.makeText(getApplicationContext(),"The multiplication of the two numbers
            is:"+mul,Toast.LENGTH_SHORT).show();
    }
});

div_btn.setOnClickListener(new View.OnClickListener(){
    @Override
    public void onClick(View view){
        String number1_text=number1.getText().toString();
        String number2_text=number2.getText().toString();
```

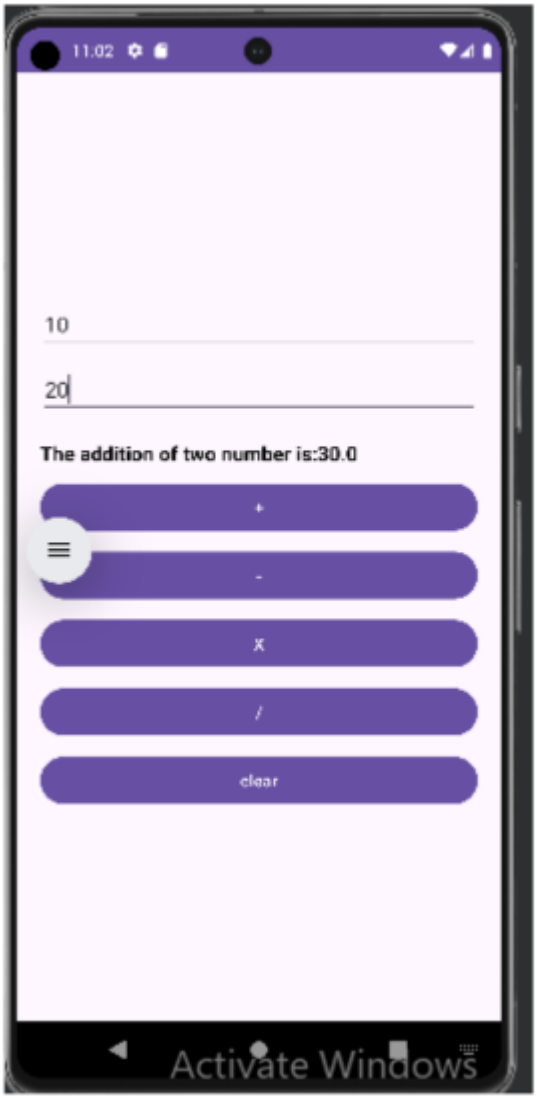
```
            int num1=Integer.parseInt(number1_text);
            int num2=Integer.parseInt(number2_text);
            float div=num1/num2;
            result_text.setText("The division of the two number is:"+div);

            Toast.makeText(getApplicationContext(),"The division of the two numbers
                is:"+div,Toast.LENGTH_SHORT).show();
        }
    });

    clear_btn.setOnClickListener(new View.OnClickListener(){
        @Override
        public void onClick(View view){
            number1.setText("");
            number2.setText("");
            result_text.setText((""));

            Toast.makeText(getApplicationContext(),"Inputs
                Cleared....",Toast.LENGTH_SHORT).show();
        }
    });
  }
}
```

Output:



Result: The program is executed successfully and output is verified.

PROGRAM – 6

Aim: Write a program to create a facebook page using RelativeLayout, Set properties using .xml file.

Program

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:orientation="vertical"
    android:layout_height="match_parent"
    android:id="@+id/mainlay"
    tools:context=".MainActivity">
    <ImageView
        android:layout_width="match_parent"
        android:layout_height="150dp"
        android:src="@drawable/fb"
        android:scaleType="centerCrop"/>
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Welcome to Facebook"
        android:textAlignment="center"
        android:layout_marginTop="5dp"
        android:layout_gravity="center"/>
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
```

```
    android:paddingHorizontal="30dp"
    android:layout_marginTop="20dp"
    android:orientation="vertical"/>
<EditText
    android:id="@+id/fb_id"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Phone or email"
    android:digits=""
    android:inputType="text"/>
<EditText
    android:id="@+id/fb_pass"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="20dp"
    android:hint="Password"
    android:inputType="textPassword"/>
<Button
    android:id="@+id/fb_loginbtn"
    android:layout_width="match_parent"
    android:layout_height="60dp"
    android:text="Log In"
    android:layout_marginTop="20dp"
    android:textAllCaps="false"
    android:backgroundTint="#3F51B5"/>
<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Forgot Password?"
```

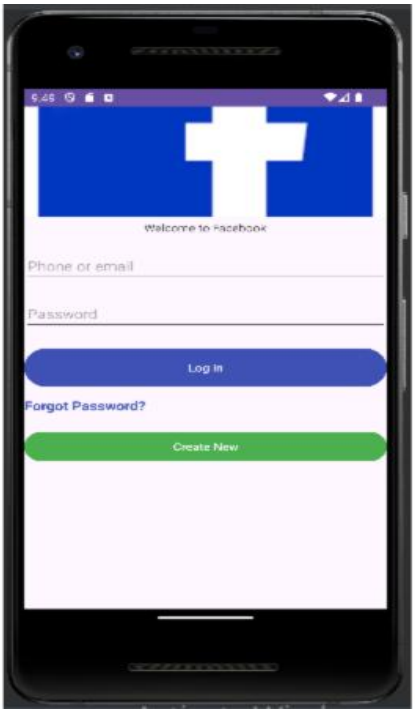
```
        android:textStyle="bold"
        android:textColor="#3F51B5"
        android:textSize="17sp"
        android:layout_marginTop="10dp"/>
    <Button
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Create New"
        android:layout_marginTop="20dp"
        android:textAllCaps="false"
        android:backgroundTint="#4CAF50"/>
</LinearLayout>
```

MainActivity.java

```
package com.example.facebook;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;

public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

Output:



Result: The program is executed successfully and output is verified.

PROGRAM – 10

Aim: Write a program to implement Intent to navigate between multiple activities.

Program

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    tools:context=".MainActivity">
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Home Page"
        android:textSize="17sp"/>
    <Button
        android:id="@+id/button"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="View Profile"
        android:layout_marginTop="1dp"/>
</LinearLayout>
```

MainActivity.java

```
package com.example.intent;
import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;
```

```
import android.os.Bundle;
import android.view.View;
import android.widget.Button;

public class MainActivity extends AppCompatActivity {
Button button;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        button=findViewById(R.id.button);

        button.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                Intent intent=new Intent(getApplicationContext(),Navigate2.class);
                startActivity(intent);
            }
        });
    }
}
```

activity_navigate.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
```

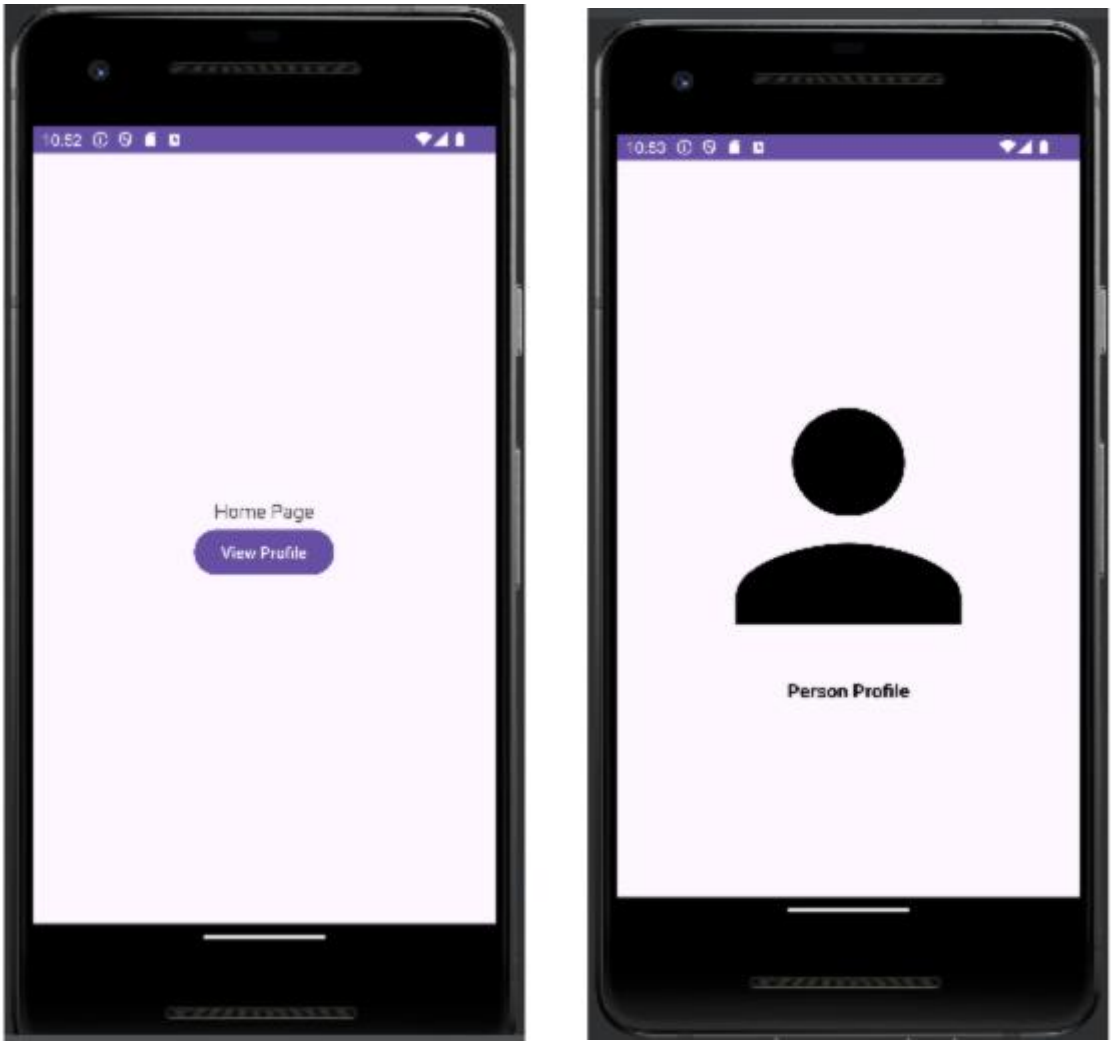
```
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    tools:context=".Navigate2">
    <ImageView
        android:layout_width="match_parent"
        android:layout_height="300dp"
        android:src="@drawable/baseline_person_24"/>
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Person Profile"
        android:textColor="@color/black"
        android:textStyle="bold"
        android:textSize="17sp"/>
</LinearLayout>
```

Navigate2.java

```
package com.example.intent;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;

public class Navigate2 extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_navigate);
    }
}
```

Output:



Result: The program is executed successfully and output is verified.

PROGRAM – 14

Aim: Write a program to develop an application that uses GridView with images and display AlertBox on selection.

Program

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    tools:context=".MainActivity">
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="My Apps"
        android:layout_marginTop="30dp"
        android:textColor="@color/black"
        android:textSize="20sp"
        android:textStyle="bold"/>
    <GridLayout
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="30dp"
        android:columnCount="3"
        android:orientation="horizontal"
        android:rowCount="3">
```

```
    <ImageView
        android:id="@+id/wifi_btn"
        android:layout_width="80dp"
        android:layout_height="80dp"
        android:padding="13dp"
        android:src="@drawable/baseline_wifi_24"
        android:textColor="@color/white"/>
    <ImageView
        android:id="@+id/bluetooth_btn"
        android:layout_width="80dp"
        android:layout_height="80dp"
        android:padding="13dp"
        android:src="@drawable/baseline_bluetooth_24"
        android:textColor="@color/white"/>
    <ImageView
        android:id="@+id/volume_btn"
        android:layout_width="80dp"
        android:layout_height="80dp"
        android:padding="13dp"
        android:src="@drawable/baseline_volume_up_24"
        android:textColor="@color/white"/>
    </GridLayout>
</LinearLayout>
```

MainActivity.java

```
package com.example.gridviewdr;
import androidx.appcompat.app.AlertDialog;
import androidx.appcompat.app.AppCompatActivity;
import android.content.DialogInterface;
```

```
import android.os.Bundle;
import android.view.View;
import android.widget.ImageButton;
import android.widget.ImageView;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {
ImageView wifi_btn,bluetooth_btn,volume_btn;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        wifi_btn=findViewById(R.id.wifi_btn);
        bluetooth_btn=findViewById(R.id.bluetooth_btn);
        volume_btn=findViewById(R.id.volume_btn);

        wifi_btn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                myAlertBox("Album Option");
            }
        });

        bluetooth_btn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                myAlertBox("Mail Option");
```

```
        }
    });

    volume_btn.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View view) {
            myAlertBox("Money Option");
        }
    });
}

public void myAlertBox(String buttonname) {
    AlertDialog.Builder builder = new AlertDialog.Builder(MainActivity.this);
    builder.setMessage(("Click OK to select option"));
    builder.setTitle("Alert!");
    builder.setCancelable(false);
    builder.setPositiveButton("Yes", new DialogInterface.OnClickListener() {

        @Override
        public void onClick(DialogInterface dialogInterface, int i) {

            Toast.makeText(getApplicationContext(),buttonname+"is successfully
                    selected",Toast.LENGTH_SHORT).show();
        }
    });

    builder.setNegativeButton("No",(DialogInterface.OnClickListener)(dialog,which)->
    {
        dialog.cancel();
```

```
        });
        AlertDialog alertDialog = builder.create();
        alertDialog.show();
    }
}
```

Output:





Result: The program is executed successfully and output is verified.

PROGRAM – 5

Aim: Write a program to design a registration activity and store registration details in local memory of phone using Intents and SharedPreferences.

Program

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:paddingHorizontal="20dp"
    android:id="@+id/main_layout"
    android:paddingVertical="10dp"
    android:gravity="center"
    tools:context=".MainActivity">
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Registration Form"
        android:textStyle="bold"
        android:textColor="@color/black"
        android:textSize="20sp"/>
    <EditText
        android:id="@+id/fullname"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:inputType="textPersonName"
```

```
        android:layout_marginTop="30dp"
        android:hint="Full Name"/>
    <EditText
        android:id="@+id/emailid"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:inputType="textEmailAddress"
        android:layout_marginTop="10dp"
        android:hint="Email ID"/>
    <EditText
        android:id="@+id/password"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:inputType="textPassword"
        android:hint="Password"/>
    <Button
        android:id="@+id/register_btn"
        android:layout_width="201dp"
        android:layout_height="wrap_content"
        android:layout_marginTop="20dp"
        android:text="Register"/>
</LinearLayout>
```

MainActivity.java

```
package com.example.registration;
import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;
import android.content.SharedPreferences;
```

```
import android.os.Bundle;
import android.util.Patterns;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.LinearLayout;
import android.widget.RadioButton;
import android.widget.RadioGroup;
import android.widget.TextView;

public class MainActivity extends AppCompatActivity {
EditText fullname,emailid,password;
Button register_btn;
LinearLayout main_layout;
    @Override
      protected void onCreate(Bundle savedInstanceState) {
          super.onCreate(savedInstanceState);
          setContentView(R.layout.activity_main);

          fullname=findViewById(R.id.fullname);
          emailid=findViewById(R.id.emailid);
          password=findViewById(R.id.password);
          register_btn=findViewById(R.id.register_btn);
          main_layout=findViewById(R.id.main_layout);

          register_btn.setOnClickListener(new View.OnClickListener(){

                @Override
                 public void onClick(View view){
```

```
fullname.setError(null);
emailid.setError(null);
password.setError(null);

String password_regex = "^(?=.*[0-9])(?=.*[a-z])(?=.*[A-
        Z])(?=.*[@#$%^&+=])(?=\\S+$).{6,}$";
String fullname_txt=fullname.getText().toString();
String emailid_txt=emailid.getText().toString();
String password_txt=password.getText().toString();

if(fullname_txt.equals("")) {
   fullname.requestFocus();
   fullname.setError("Please enter fullname!!");
} else if(emailid_txt.equals("")) {
   emailid.requestFocus();
   emailid.setError("Please enter emailid_id!!");
} else if(!Patterns.EMAIL_ADDRESS.matcher(emailid_txt).matches()) {
   emailid.requestFocus();
   emailid.setError("Please enter a valid emailid_id!!");
} else if(!password_txt.matches(password_regex)){
   password.requestFocus();
   password.setError("Password should contain -\n a digit must occur atleast once
  \n a lower case lettre must occur atleast \n an uppercase letter occur atleast once
  \n a special character like @#$%^&+=\n No blank spaces allowed \n atleast 6
  character");
} else {
   SharedPreferences  pref= getSharedPreferences("register_data",
        MODE_PRIVATE);
   SharedPreferences.Editor pref_edit=pref.edit();
   pref_edit.putString("reg_fullname",fullname_txt);
```

```
                pref_edit.putString("reg_emailid",emailid_txt);
                pref_edit.putString("reg_password",password_txt);
                pref_edit.apply();
                Intent intent=new Intent(getApplicationContext(),IntendS2.class);
                startActivity(intent);
            }
        }
    });
  }
}
```

activity_intend_s2.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:app="http://schemas.android.com/apk/res-auto"
   xmlns:tools="http://schemas.android.com/tools"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:gravity="center"
   android:padding="10dp"
   tools:context=".IntendS2">
   <TextView
      android:id="@+id/fullname_result"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:layout_marginTop="10dp"/>
  <TextView
      android:id="@+id/emailid_result"
```

```
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="10dp"/>
    <TextView
        android:id="@+id/password_result"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="10dp"/>
</LinearLayout>
```

IntendS2.java

```
package com.example.registration;
import androidx.appcompat.app.AppCompatActivity;
import android.content.SharedPreferences;
import android.os.Bundle;
import android.widget.TextView;

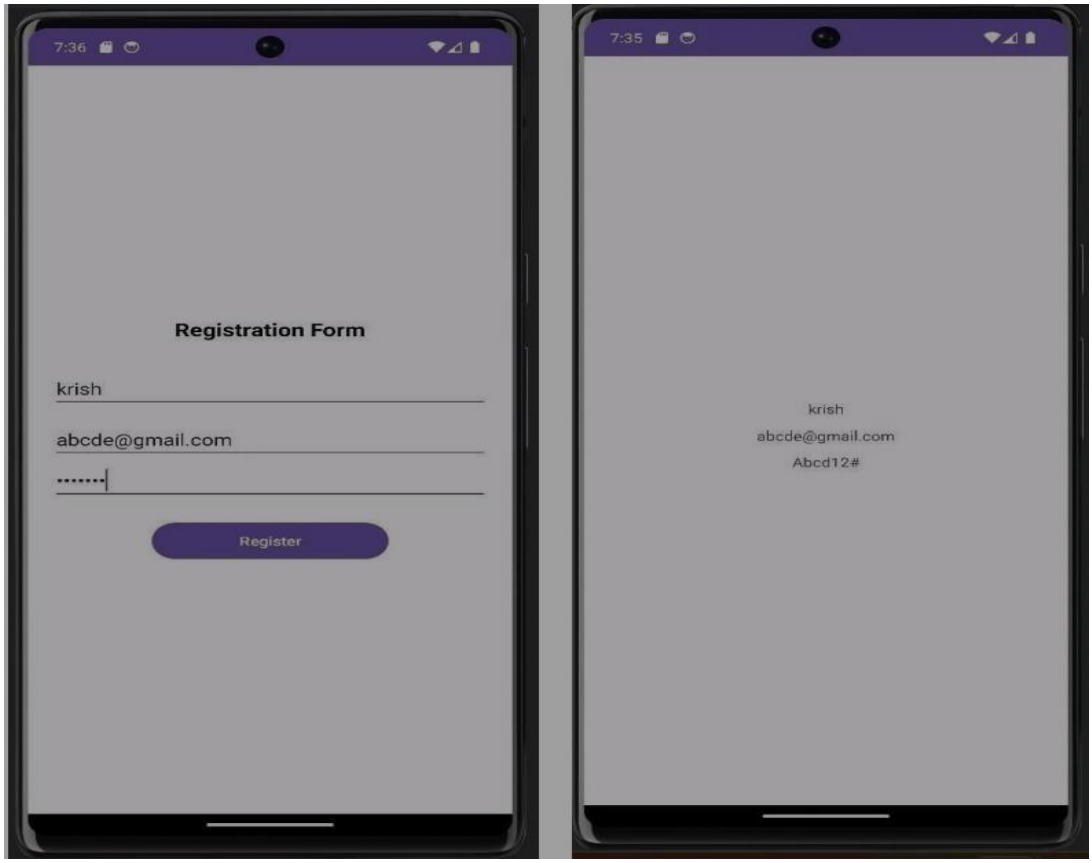
public class IntendS2 extends AppCompatActivity {
TextView fullname_result, emailid_result, password_result;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_intend_s2);

        fullname_result = findViewById(R.id.fullname_result);
        emailid_result = findViewById(R.id.emailid_result);
        password_result = findViewById(R.id.password_result);
        SharedPreferences pref = getSharedPreferences("register_data", MODE_PRIVATE);
```

```
        String name = pref.getString("reg_fullname", "NotAvailable!!");
        String email = pref.getString("reg_emailid", "NotAvailable!!");
        String password = pref.getString("reg_password", "NotAvailable!!");
        fullname_result.setText(name);
        emailid_result.setText(email);
        password_result.setText(password);
    }
}
```

Output:



Result: The program is executed successfully and output is verified.